

Physics 513 QFT Midterm

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The questions are in blue and my answers are in black. Because I am auditing the class and not competing for a grade I will use my solutions to solve problem 1.

1. 1

Consider the extension of the proper Lorentz group to include space reflection or parity P . If the position is $\vec{x} = (t, x, y, z)$ then parity is defined by $P\vec{x} = (t, -x, -y, -z)$. Positions are not actually vectors but in flat spacetime we can pretend that they are (just like in Newtonian mechanics).

1.1. 1.a

Show that P commutes with the rotation generators J_i and anticommutes with the boost generators K_i .

This problem is best solved in matrix, not index notation. The parity matrix is

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (1)$$

I derived the boosts and rotations in HW. 1.

$$\begin{aligned} J_x &= i \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} & J_y &= i \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \\ J_z &= i \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} & K_x &= i \begin{pmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\ K_y &= i \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} & K_z &= i \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}. \end{aligned} \quad (2)$$

Then using Mathematica I confirm that

$$[P, J_i] = \{P, K_j\} = 0. \quad (3)$$

1.2. 1.b

Show that there is no 2×2 matrix with the properties of P within the $(1/2, 0)$ representation of the Lorentz group. Therefore the $(1/2, 0)$ representation of the Lorentz group cannot be extended to include parity.

The irreducible representation of the Lorentz group is

$$\mathbf{J}^\pm = \frac{\mathbf{J} \pm i\mathbf{K}}{2}, \quad (4)$$

where I use bold face for 3-vectors. Their sum is

$$\mathbf{J}^+ + \mathbf{J}^- = \mathbf{J}. \quad (5)$$

This is the same as adding two angular momenta in non-relativistic quantum mechanics. The different irreducible representations are labeled by the total angular momenta quantum numbers (j^+, j^-) .

In the $(1/2, 0)$ representation the basis states are

$$|j^+ j^- m^+ m^-\rangle = \left| \frac{1}{2} 0 m 0 \right\rangle = |m\rangle \quad (6)$$

where I suppress all quantum numbers except $m^+ = m$ because they are constants. This representation is 2D

$$(2j^+ + 1)(2j^- + 1) = (2j^+ + 1)1 = \frac{2}{2} + 1 = 1 + 1 = 2. \quad (7)$$

The basis states are

$$|+\rangle = (1, 0), \quad |-\rangle = (0, 1). \quad (8)$$

I construct the angular momentum operators for the $+$ and $-$ representations.

$$J_z^- |+\rangle = J_z^- |-\rangle = 0. \quad (9)$$

$$\boxed{J_x^- = J_y^- = J_z^- = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}} \quad (10)$$

The $\mathbf{J}^+$ vector operator is the same as the spin-1/2 vector operator, so I can write down the answer immediately.

$$\boxed{J_i^+ = \frac{\sigma_i}{2}}, \quad (11)$$

where the Pauli matrices are

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = i \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \sigma_z = i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (12)$$

I calculate the rotation generators.

$$\mathbf{J}^\pm = \frac{\mathbf{J} \pm i\mathbf{K}}{2} \quad (13)$$

$$\mathbf{J}^+ + \mathbf{J}^- = \frac{\mathbf{J} + i\mathbf{K} + \mathbf{J} - i\mathbf{K}}{2} = \mathbf{J} \quad (14)$$

$$\mathbf{J}^+ + \cancel{\mathbf{J}^-} = \mathbf{J} \implies \mathbf{J} = \mathbf{J}^+ \quad (15)$$

$$\boxed{J_i = \frac{\sigma_i}{2}} \quad (16)$$

I calculate the boost generators.

$$\mathbf{J}^+ - \cancel{\mathbf{J}^-} = i\mathbf{K} \implies i\mathbf{K} = \mathbf{J}^+ = \mathbf{J} \implies \mathbf{K} = -i\mathbf{J} \quad (17)$$

$$\boxed{K_j = -\frac{i}{2}\sigma_j} \quad (18)$$

I assume that the parity matrix can exist. This matrix must satisfy $[P, J_i] = 0$.

$$[P, J_i] = \left[P, \frac{\sigma_i}{2} \right] = \frac{[P, \sigma_i]}{2} \quad (19)$$

$$[P, J_i] = 0 \implies [P, \sigma_i] = 0 \quad (20)$$

It must also satisfy $\{P, K_i\} = 0$.

$$\{P, K_i\} = \left\{ P, -\frac{i}{2}\sigma_i \right\} = -\frac{i}{2}\{P, \sigma_i\} \quad (21)$$

$$\{P, K_i\} = 0 \implies \{P, \sigma_i\} = 0. \quad (22)$$

No nonzero matrix can satisfy

$$\{P, \sigma_i\} = [P, \sigma_i] = 0. \quad (23)$$

Therefore the parity operation does not exist in the $(1/2, 0)$ representation. QED.

2. 2

Consider the free charged scalar field described by the Lagrangian

$$\mathcal{L} = -(\partial_\mu \phi^\dagger)(\partial^\mu \phi) - m^2 \phi^\dagger \phi. \quad (24)$$

In this theory the charged current is given by

$$iJ^\mu = \phi^\dagger \partial^\mu \phi - \phi \partial^\mu \phi^\dagger. \quad (25)$$

Find an expression for the total charge operator in terms of the creation and annihilation operators of the particles and anti-particles.

The field operators are

$$\phi(\vec{x}) = \int d^3p \frac{a_{\mathbf{p}} e^{-i\vec{p} \cdot \vec{x}} + b_{\mathbf{p}}^\dagger e^{i\vec{p} \cdot \vec{x}}}{(2\pi)^3 \sqrt{2E_p}}, \quad (26)$$

$$\phi^\dagger(\vec{x}) = \int d^3p \frac{a_{\mathbf{p}}^\dagger e^{i\vec{p} \cdot \vec{x}} + b_{\mathbf{p}} e^{-i\vec{p} \cdot \vec{x}}}{(2\pi)^3 \sqrt{2E_p}}, \quad (27)$$

$$(28)$$

where $E_p = \sqrt{p^2 + m^2}$ where $p^2 = (p^1)^2 + (p^2)^2 + (p^3)^2$.

$$iJ^\mu = \eta^{\mu\beta} \phi^\dagger \partial_\beta \phi - \eta^{\mu\beta} \phi \partial_\beta \phi^\dagger \quad (29)$$

The inverse metric in Cartesian coordinates, (t, x, y, z) , is

$$\eta^{\alpha\beta} = \text{diag}(1, -1, -1, -1) \quad (30)$$

The charge density is

$$\rho = j^0. \quad (31)$$

$$iJ^0 = \eta^{0\beta} \phi^\dagger \partial_\beta \phi - \eta^{0\beta} \phi \partial_\beta \phi^\dagger \quad (32)$$

$$i\rho = \eta^{00} \phi^\dagger \partial_0 \phi - \eta^{00} \phi \partial_0 \phi^\dagger \quad (33)$$

$$i\rho = \phi^\dagger \partial_0 \phi - \phi \partial_0 \phi^\dagger \quad (34)$$

$$\partial_0 \phi = \dot{\phi} \quad (35)$$

$$i\rho = \phi^\dagger \dot{\phi} - \phi \dot{\phi}^\dagger \quad (36)$$

$$i\rho(\vec{x}) = \phi^\dagger(\vec{x}) \dot{\phi}(\vec{x}) - \phi(\vec{x}) \dot{\phi}^\dagger(\vec{x}) \quad (37)$$

$$\pi(\vec{x}) = \dot{\phi}(\vec{x}), \quad \pi^\dagger(\vec{x}) = \dot{\phi}^\dagger(\vec{x}) \quad (38)$$

$$\pi(\vec{x}) = \int d^3p \frac{a_{\mathbf{p}} \partial_t e^{-i\vec{p} \cdot \vec{x}} + b_{\mathbf{p}}^\dagger \partial_t e^{i\vec{p} \cdot \vec{x}}}{(2\pi)^3 \sqrt{2E_p}} \quad (39)$$

$$\vec{p} = (E_p, \mathbf{p}), \quad \vec{x} = (t, \mathbf{x}) \quad (40)$$

$$\vec{p} \cdot \vec{x} = (E_p, \mathbf{p}) \cdot (t, \mathbf{x}) = E_p t - \mathbf{p} \cdot \mathbf{x} \quad (41)$$

$$\pi(\vec{x}) = \int d^3p \frac{a_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}} \partial_t e^{-iE_p t} + b_{\mathbf{p}}^\dagger e^{-i\mathbf{p}\cdot\mathbf{x}} \partial_t e^{iE_p t}}{(2\pi)^3 \sqrt{2E_p}} \quad (42)$$

$$\pi(\vec{x}) = \int d^3p \frac{a_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}} (-iE_p) e^{-iE_p t} + b_{\mathbf{p}}^\dagger e^{-i\mathbf{p}\cdot\mathbf{x}} (iE_p) e^{iE_p t}}{(2\pi)^3 \sqrt{2E_p}} \quad (43)$$

$$\pi(\vec{x}) = i \int d^3p E_p \frac{b_{\mathbf{p}}^\dagger e^{-i\mathbf{p}\cdot\mathbf{x}} e^{iE_p t} - a_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}} e^{-iE_p t}}{(2\pi)^3 \sqrt{2E_p}} \quad (44)$$

$$\pi(\vec{x}) = i \int d^3p \sqrt{\frac{E_p}{2}} \frac{b_{\mathbf{p}}^\dagger e^{i(E_p t - \mathbf{p}\cdot\mathbf{x})} - a_{\mathbf{p}} e^{-i(E_p t - \mathbf{p}\cdot\mathbf{x})}}{(2\pi)^3} \quad (45)$$

$$\pi(\vec{x}) = i \int d^3p \sqrt{\frac{E_p}{2}} \frac{b_{\mathbf{p}}^\dagger e^{i\vec{p}\cdot\vec{x}} - a_{\mathbf{p}} e^{-i\vec{p}\cdot\vec{x}}}{(2\pi)^3} \quad (46)$$

$$\pi^\dagger(\vec{x}) = i \int d^3p \sqrt{\frac{E_p}{2}} \frac{a_{\mathbf{p}}^\dagger e^{i\vec{p}\cdot\vec{x}} - b_{\mathbf{p}} e^{-i\vec{p}\cdot\vec{x}}}{(2\pi)^3} \quad (47)$$

$$i\rho(\vec{x}) = \phi^\dagger(\vec{x})\pi(\vec{x}) - \phi(\vec{x})\pi^\dagger(\vec{x}) \quad (48)$$

$$\begin{aligned} i\rho(\vec{x}) = & \int d^3p \frac{a_{\mathbf{p}}^\dagger e^{i\vec{p}\cdot\vec{x}} + b_{\mathbf{p}} e^{-i\vec{p}\cdot\vec{x}}}{(2\pi)^3 \sqrt{2E_p}} i \int d^3k \sqrt{\frac{E_k}{2}} \frac{b_{\mathbf{k}}^\dagger e^{i\vec{k}\cdot\vec{x}} - a_{\mathbf{k}} e^{-i\vec{k}\cdot\vec{x}}}{(2\pi)^3} - \\ & \int d^3p \frac{a_{\mathbf{p}} e^{-i\vec{p}\cdot\vec{x}} + b_{\mathbf{p}}^\dagger e^{i\vec{p}\cdot\vec{x}}}{(2\pi)^3 \sqrt{2E_p}} i \int d^3k \sqrt{\frac{E_k}{2}} \frac{a_{\mathbf{k}}^\dagger e^{i\vec{k}\cdot\vec{x}} - b_{\mathbf{k}} e^{-i\vec{k}\cdot\vec{x}}}{(2\pi)^3} \end{aligned} \quad (49)$$

$$\begin{aligned} \rho(\vec{x}) = & \int \frac{d^3p d^3k}{2(2\pi)^6} \sqrt{\frac{E_k}{E_p}} (a_{\mathbf{p}}^\dagger e^{i\vec{p}\cdot\vec{x}} + b_{\mathbf{p}} e^{-i\vec{p}\cdot\vec{x}}) (b_{\mathbf{k}}^\dagger e^{i\vec{k}\cdot\vec{x}} - a_{\mathbf{k}} e^{-i\vec{k}\cdot\vec{x}}) \\ & - \int \frac{d^3p d^3k}{2(2\pi)^6} \sqrt{\frac{E_k}{E_p}} (a_{\mathbf{p}} e^{-i\vec{p}\cdot\vec{x}} + b_{\mathbf{p}}^\dagger e^{i\vec{p}\cdot\vec{x}}) (a_{\mathbf{k}}^\dagger e^{i\vec{k}\cdot\vec{x}} - b_{\mathbf{k}} e^{-i\vec{k}\cdot\vec{x}}) \end{aligned} \quad (50)$$

$$\begin{aligned} \rho(\vec{x}) = & \int \frac{d^3p d^3k}{2(2\pi)^6} \sqrt{\frac{E_k}{E_p}} \\ & [(a_{\mathbf{p}}^\dagger e^{i\vec{p}\cdot\vec{x}} + b_{\mathbf{p}} e^{-i\vec{p}\cdot\vec{x}}) (b_{\mathbf{k}}^\dagger e^{i\vec{k}\cdot\vec{x}} - a_{\mathbf{k}} e^{-i\vec{k}\cdot\vec{x}}) \\ & - (a_{\mathbf{p}} e^{-i\vec{p}\cdot\vec{x}} + b_{\mathbf{p}}^\dagger e^{i\vec{p}\cdot\vec{x}}) (a_{\mathbf{k}}^\dagger e^{i\vec{k}\cdot\vec{x}} - b_{\mathbf{k}} e^{-i\vec{k}\cdot\vec{x}})] \end{aligned} \quad (51)$$

$$\begin{aligned} \rho(\vec{x}) = & \int \frac{d^3p d^3k}{2(2\pi)^6} \sqrt{\frac{E_k}{E_p}} [a_{\mathbf{p}}^\dagger b_{\mathbf{k}}^\dagger e^{i(\vec{p}+\vec{k})\cdot\vec{x}} - a_{\mathbf{p}}^\dagger a_{\mathbf{k}} e^{i(\vec{p}-\vec{k})\cdot\vec{x}} \\ & + b_{\mathbf{p}} b_{\mathbf{k}}^\dagger e^{i(\vec{k}-\vec{p})\cdot\vec{x}} - b_{\mathbf{p}} a_{\mathbf{k}} e^{-i(\vec{p}+\vec{k})\cdot\vec{x}} - a_{\mathbf{p}} a_{\mathbf{k}}^\dagger e^{i(\vec{k}-\vec{p})\cdot\vec{x}} \\ & + a_{\mathbf{p}} b_{\mathbf{k}} e^{-i(\vec{p}+\vec{k})\cdot\vec{x}} - b_{\mathbf{p}}^\dagger a_{\mathbf{k}}^\dagger e^{i(\vec{p}+\vec{k})\cdot\vec{x}} + b_{\mathbf{p}}^\dagger b_{\mathbf{k}} e^{i(\vec{p}-\vec{k})\cdot\vec{x}}] \end{aligned} \quad (52)$$

$$\begin{aligned} \rho(\vec{x}) = & \int \frac{d^3p d^3k}{2(2\pi)^6} \sqrt{\frac{E_k}{E_p}} [(a_{\mathbf{p}}^\dagger b_{\mathbf{k}}^\dagger - b_{\mathbf{p}}^\dagger a_{\mathbf{k}}^\dagger) e^{i(\vec{k}+\vec{p})\cdot\vec{x}} \\ & + (b_{\mathbf{p}} b_{\mathbf{k}}^\dagger - a_{\mathbf{p}} a_{\mathbf{k}}^\dagger) e^{i(\vec{k}-\vec{p})\cdot\vec{x}} + (a_{\mathbf{p}} b_{\mathbf{k}} - b_{\mathbf{p}} a_{\mathbf{k}}) e^{-i(\vec{p}+\vec{k})\cdot\vec{x}} \\ & + (b_{\mathbf{p}}^\dagger b_{\mathbf{k}} - a_{\mathbf{p}}^\dagger a_{\mathbf{k}}) e^{i(\vec{p}-\vec{k})\cdot\vec{x}}] \end{aligned} \quad (53)$$

$$\vec{v}_\pm = \vec{p} \pm \vec{k} = (E_{v_\pm}, \mathbf{v}_\pm) = (E_p \pm E_k, \mathbf{p} \pm \mathbf{k}), \quad (54)$$

$$\begin{aligned} \rho(\vec{x}) = & \int \frac{d^3p d^3k}{2(2\pi)^6} \sqrt{\frac{E_k}{E_p}} [(a_{\mathbf{p}}^\dagger b_{\mathbf{k}}^\dagger - b_{\mathbf{p}}^\dagger a_{\mathbf{k}}^\dagger) e^{i\vec{v}_+\cdot\vec{x}} \\ & + (b_{\mathbf{p}} b_{\mathbf{k}}^\dagger - a_{\mathbf{p}} a_{\mathbf{k}}^\dagger) e^{-i\vec{v}_-\cdot\vec{x}} + (a_{\mathbf{p}} b_{\mathbf{k}} - b_{\mathbf{p}} a_{\mathbf{k}}) e^{-i\vec{v}_+\cdot\vec{x}} \\ & + (b_{\mathbf{p}}^\dagger b_{\mathbf{k}} - a_{\mathbf{p}}^\dagger a_{\mathbf{k}}) e^{i\vec{v}_-\cdot\vec{x}}] \end{aligned} \quad (55)$$

$$\begin{aligned} \rho(\vec{x}) = & \int \frac{d^3p d^3k}{2(2\pi)^6} \sqrt{\frac{E_k}{E_p}} [(a_{\mathbf{p}}^\dagger b_{\mathbf{k}}^\dagger - b_{\mathbf{p}}^\dagger a_{\mathbf{k}}^\dagger) e^{itE_{v_+}} e^{-i\mathbf{v}_+\cdot\mathbf{x}} \\ & + (b_{\mathbf{p}} b_{\mathbf{k}}^\dagger - a_{\mathbf{p}} a_{\mathbf{k}}^\dagger) e^{-itE_{v_-}} e^{i\mathbf{v}_-\cdot\mathbf{x}} + (a_{\mathbf{p}} b_{\mathbf{k}} - b_{\mathbf{p}} a_{\mathbf{k}}) e^{-itE_{v_+}} e^{i\mathbf{v}_+\cdot\mathbf{x}} \\ & + (b_{\mathbf{p}}^\dagger b_{\mathbf{k}} - a_{\mathbf{p}}^\dagger a_{\mathbf{k}}) e^{itE_{v_-}} e^{-i\mathbf{v}_-\cdot\mathbf{x}}] \end{aligned} \quad (56)$$

The total charge is

$$Q = \int d^3x \rho. \quad (57)$$

$$\begin{aligned} Q = & \int \frac{d^3p d^3k}{2(2\pi)^6} \sqrt{\frac{E_k}{E_p}} \\ & \left[(a_{\mathbf{p}}^\dagger b_{\mathbf{k}}^\dagger - b_{\mathbf{p}}^\dagger a_{\mathbf{k}}^\dagger) e^{itE_{v_+}} \int d^3x e^{-i\mathbf{v}_+\cdot\mathbf{x}} \right. \\ & + (b_{\mathbf{p}} b_{\mathbf{k}}^\dagger - a_{\mathbf{p}} a_{\mathbf{k}}^\dagger) e^{-itE_{v_-}} \int d^3x e^{i\mathbf{v}_-\cdot\mathbf{x}} + \\ & (a_{\mathbf{p}} b_{\mathbf{k}} - b_{\mathbf{p}} a_{\mathbf{k}}) e^{-itE_{v_+}} \int d^3x e^{i\mathbf{v}_+\cdot\mathbf{x}} \\ & \left. + (b_{\mathbf{p}}^\dagger b_{\mathbf{k}} - a_{\mathbf{p}}^\dagger a_{\mathbf{k}}) e^{itE_{v_-}} \int d^3x e^{-i\mathbf{v}_-\cdot\mathbf{x}} \right] \end{aligned} \quad (58)$$

$$\int d^3x e^{i\mathbf{p}\cdot\mathbf{x}} = (2\pi)^3 \delta^{(3)}(\mathbf{p}) \quad (59)$$

$$\begin{aligned}
 Q = \int \frac{d^3p d^3k}{2(2\pi)^3} \sqrt{\frac{E_k}{E_p}} & \\
 & [(a_{\mathbf{p}}^\dagger b_{\mathbf{k}}^\dagger - b_{\mathbf{p}}^\dagger a_{\mathbf{k}}^\dagger) e^{itE_{v+}} \delta^{(3)}(-\mathbf{v}_+) \\
 & + (b_{\mathbf{p}} b_{\mathbf{k}}^\dagger - a_{\mathbf{p}} a_{\mathbf{k}}^\dagger) e^{-itE_{v-}} \delta^{(3)}(\mathbf{v}_-) + \\
 & (a_{\mathbf{p}} b_{\mathbf{k}} - b_{\mathbf{p}} a_{\mathbf{k}}) e^{-itE_{v+}} \delta^{(3)}(\mathbf{v}_+) \\
 & + (b_{\mathbf{p}}^\dagger b_{\mathbf{k}} - a_{\mathbf{p}}^\dagger a_{\mathbf{k}}) e^{itE_{v-}} \delta^{(3)}(-\mathbf{v}_-)]
 \end{aligned} \tag{60}$$

$$\delta(\mathbf{p}) = \delta(-\mathbf{p}) \tag{61}$$

$$\begin{aligned}
 Q = \int \frac{d^3p d^3k}{2(2\pi)^3} \sqrt{\frac{E_k}{E_p}} & [(a_{\mathbf{p}}^\dagger b_{\mathbf{k}}^\dagger - b_{\mathbf{p}}^\dagger a_{\mathbf{k}}^\dagger) e^{itE_{v+}} \\
 & + (a_{\mathbf{p}} b_{\mathbf{k}} - b_{\mathbf{p}} a_{\mathbf{k}}) e^{-itE_{v+}}] \delta^{(3)}(\mathbf{v}_+) \\
 & + \int \frac{d^3p d^3k}{2(2\pi)^3} \sqrt{\frac{E_k}{E_p}} [(b_{\mathbf{p}} b_{\mathbf{k}}^\dagger - a_{\mathbf{p}} a_{\mathbf{k}}^\dagger) e^{-itE_{v-}} \\
 & + (b_{\mathbf{p}}^\dagger b_{\mathbf{k}} - a_{\mathbf{p}}^\dagger a_{\mathbf{k}}) e^{itE_{v-}}] \delta^{(3)}(\mathbf{v}_-)
 \end{aligned} \tag{62}$$

$$\begin{aligned}
 Q = \int \frac{d^3p d^3k}{2(2\pi)^3} \sqrt{\frac{E_k}{E_p}} & [(a_{\mathbf{p}}^\dagger b_{\mathbf{k}}^\dagger - b_{\mathbf{p}}^\dagger a_{\mathbf{k}}^\dagger) e^{it(E_p+E_k)} \\
 & + (a_{\mathbf{p}} b_{\mathbf{k}} - b_{\mathbf{p}} a_{\mathbf{k}}) e^{-it(E_p+E_k)}] \delta^{(3)}(\mathbf{p}+\mathbf{k}) \\
 & + \int \frac{d^3p d^3k}{2(2\pi)^3} \sqrt{\frac{E_k}{E_p}} [(b_{\mathbf{p}} b_{\mathbf{k}}^\dagger - a_{\mathbf{p}} a_{\mathbf{k}}^\dagger) e^{-it(E_p-E_k)} \\
 & + (b_{\mathbf{p}}^\dagger b_{\mathbf{k}} - a_{\mathbf{p}}^\dagger a_{\mathbf{k}}) e^{it(E_p-E_k)}] \delta^{(3)}(\mathbf{p}-\mathbf{k})
 \end{aligned} \tag{63}$$

$$\begin{aligned}
 Q = \int \frac{d^3p}{2(2\pi)^3} & [(a_{\mathbf{p}}^\dagger b_{-\mathbf{p}}^\dagger - b_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger) e^{2itE_p} \\
 & + (a_{\mathbf{p}} b_{-\mathbf{p}} - b_{\mathbf{p}} a_{-\mathbf{p}}) e^{-2itE_p}] \\
 & + \int d^3p \frac{b_{\mathbf{p}} b_{\mathbf{p}}^\dagger + b_{\mathbf{p}}^\dagger b_{\mathbf{p}} - a_{\mathbf{p}} a_{\mathbf{p}}^\dagger - a_{\mathbf{p}}^\dagger a_{\mathbf{p}}}{2(2\pi)^3}
 \end{aligned} \tag{64}$$

The only nonzero commutators are

$$[a_{\mathbf{p}}, a_{\mathbf{k}}^\dagger] = [b_{\mathbf{p}}, b_{\mathbf{k}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{p}-\mathbf{k}). \tag{65}$$

$$\begin{aligned}
 Q = \int \frac{d^3p}{2(2\pi)^3} & [(a_{\mathbf{p}}^\dagger b_{-\mathbf{p}}^\dagger - b_{\mathbf{p}}^\dagger a_{-\mathbf{p}}^\dagger) e^{2itE_p} \\
 & + (a_{\mathbf{p}} b_{-\mathbf{p}} - b_{\mathbf{p}} a_{-\mathbf{p}}) e^{-2itE_p}] \\
 & + \int d^3p \frac{\{b_{\mathbf{p}}, b_{\mathbf{p}}^\dagger\} - \{a_{\mathbf{p}}, a_{\mathbf{p}}^\dagger\}}{2(2\pi)^3}
 \end{aligned} \tag{66}$$

I substitute $\mathbf{p} \rightarrow -\mathbf{p}$ in two terms.

$$\begin{aligned}
 Q = \int \frac{d^3p}{2(2\pi)^3} & [(a_{\mathbf{p}}^\dagger b_{-\mathbf{p}}^\dagger - b_{-\mathbf{p}}^\dagger a_{\mathbf{p}}^\dagger) e^{2itE_p} \\
 & + (a_{\mathbf{p}} b_{-\mathbf{p}} - b_{-\mathbf{p}} a_{\mathbf{p}}) e^{-2itE_p}] \\
 & + \int d^3p \frac{\{b_{\mathbf{p}}, b_{\mathbf{p}}^\dagger\} - \{a_{\mathbf{p}}, a_{\mathbf{p}}^\dagger\}}{2(2\pi)^3}
 \end{aligned} \tag{67}$$

$$\begin{aligned}
 Q = \int d^3p \frac{[a_{\mathbf{p}}^\dagger, b_{-\mathbf{p}}^\dagger] e^{2itE_p} + [a_{\mathbf{p}}, b_{-\mathbf{p}}] e^{-2itE_p}}{2(2\pi)^3} \\
 + \int d^3p \frac{\{b_{\mathbf{p}}, b_{\mathbf{p}}^\dagger\} - \{a_{\mathbf{p}}, a_{\mathbf{p}}^\dagger\}}{2(2\pi)^3}
 \end{aligned} \tag{68}$$

$$2(2\pi)^3 Q = \int d^3p (\{b_{\mathbf{p}}, b_{\mathbf{p}}^\dagger\} - \{a_{\mathbf{p}}, a_{\mathbf{p}}^\dagger\}) \tag{69}$$

$$2(2\pi)^3 Q = \int d^3p ([b_{\mathbf{p}}, b_{\mathbf{p}}^\dagger] - [a_{\mathbf{p}}, a_{\mathbf{p}}^\dagger] + 2b_{\mathbf{p}}^\dagger b_{\mathbf{p}} - 2a_{\mathbf{p}}^\dagger a_{\mathbf{p}}) \tag{70}$$

$$\begin{aligned}
 2(2\pi)^3 Q = \\
 \int d^3p [(2\pi)^3 \delta^{(3)}(\mathbf{0}) - (2\pi)^3 \delta^{(3)}(\mathbf{0}) + 2b_{\mathbf{p}}^\dagger b_{\mathbf{p}} - 2a_{\mathbf{p}}^\dagger a_{\mathbf{p}}]
 \end{aligned} \tag{71}$$

$$\boxed{(2\pi)^3 Q = \int d^3p (b_{\mathbf{p}}^\dagger b_{\mathbf{p}} - a_{\mathbf{p}}^\dagger a_{\mathbf{p}})} \tag{72}$$